$$= -\frac{1}{4}\left[ \log(s^2+8^2) - \log(s^2+4) \right]_s^\infty$$

$$= -\frac{1}{4}\left[ \log\left(\frac{s^2+8^2}{s^2+4}\right) \right]_s^\infty$$

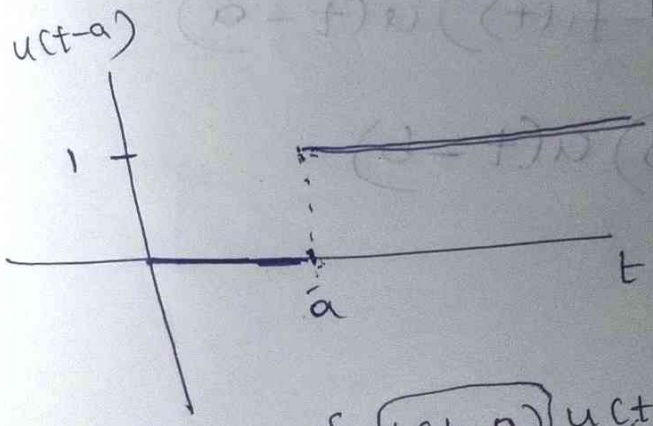
$$= -\frac{1}{4}\left[ 0 - \log\left(\frac{s^2+8^2}{s^2+4}\right) \right]$$

$$= \frac{1}{4}\log\left(\frac{s^2+8^2}{s^2+4}\right)$$

---

$u(t-a)$ | $H(t-a)$

$$= \begin{cases} 0 & t<a \quad (\text{off}) \\ 1 & t>a. \quad (\text{ON}) \end{cases}$$




$$\mathcal{L}\{u(t-a)\} = \frac{e^{-as}}{s}$$

$f(t-a) = 1$

$= \frac{1}{s}$

$$\mathcal{L}\{f(t-a)\,u(t-a)\} = e^{-as}F(s)$$

$F(s) = \mathcal{L}\{f(t)\}$

① $\mathcal{L}\{t^2 + 5(t-2)\}$

$f(t-2) = t^2$

$t \to t+2$

$f(t) = (t+2)^2 = t^2+4+4t = \frac{2}{s^3} + \frac{4}{s}$

$= e^{-2s}\left(\frac{2}{s^3} + \frac{4}{s}\right) + \frac{4}{s^2}$ Ans

$$f(t) = \begin{cases} \sin t = f_1 & 0 < t < \pi \\ \sin 2t = f_2 & \pi < t < 2\pi \\ \sin 3t = f_3 & t > 2\pi \end{cases}$$

$$f(t) = \sin t + (\sin 2t - \sin t)\,u(t-\pi) + (\sin 3t - \sin 2t)\,u(t-2\pi)$$

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{\sin t\} + \mathcal{L}\{(\sin 2t - \sin t)\,u(t-\pi)\} + \mathcal{L}\{(\sin 3t - \sin 2t)\,u(t-2\pi)\}$$

$$F(s) = \frac{1}{s^2+1} + e^{-\pi s}\left[\frac{2}{s^2+4} + \frac{1}{s^2+1}\right] + e^{-2\pi s}\left[\frac{3}{s^2+9} - \frac{2}{s^2+4}\right]$$

①

$g(t-\pi) = \sin 2t - \sin t$

$t \to t+\pi$

$g(t) = \sin 2(t+\pi) - \sin(t+\pi)$

$= \sin(2t+2\pi) - \sin(t+\pi)$

$= \sin 2t + \sin t$

$= \frac{2}{s^2+4} + \frac{1}{s^2+1}$

②

$h(t-2\pi) = \sin 3t - \sin 2t$

$t \to t+2\pi$

$h(t) = \sin(3t+6\pi) - \sin(2t+4\pi)$

$= \sin 3t - \sin 2t$

$= \frac{3}{s^2+9} - \frac{2}{s^2+4}$

Periodic fun?

$f(t)$ T

$f(t+T) = f(t)$

$$\mathcal{L}\{f(t)\} = \frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t)\,dt$$

Period.

$$f(t) = \begin{cases} E\sin\omega t & 0 < t < \pi/\omega \\ 0 & \pi/\omega < t < \frac{2\pi}{\omega} \end{cases}$$

$f(t+\frac{2\pi}{\omega}) = f(t)$

Given: $T = \frac{2\pi}{\omega}$

$$\mathcal{L}\{f(t)\} = \frac{1}{1-e^{-s(\frac{2\pi}{\omega})}} \int_0^{2\pi/\omega} e^{-st} f(t)\,dt$$

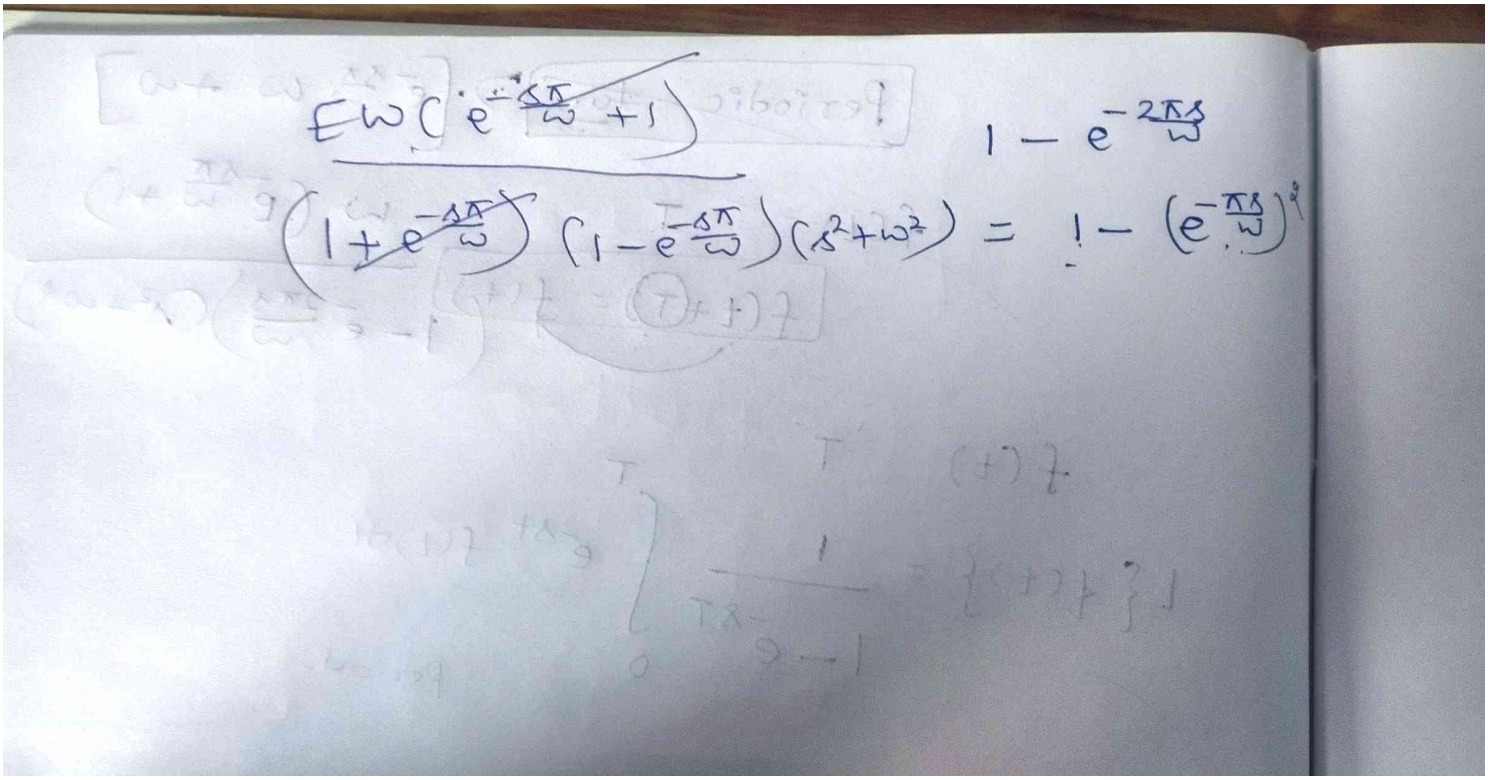
$$= \frac{1}{1-e^{-s(\frac{2\pi}{\omega})}} \left[\int_0^{\pi/\omega} e^{-st} E\sin\omega t\,dt + \int_{\pi/\omega}^{\frac{2\pi}{\omega}} e^{-st}(0)\,dt\right]$$

$x = t$, $a = -s$, $b = \omega$

$$= \frac{E}{1-e^{-s(\frac{2\pi}{\omega})}} \left[\frac{e^{-st}[-s\sin\omega t - \omega\cos\omega t]}{(-s)^2+\omega^2}\right]_0^{\pi/\omega}$$

$$= \frac{E}{(1-e^{-s(\frac{2\pi}{\omega})})(s^2+\omega^2)} \left[e^{-s\frac{\pi}{\omega}}\left[-s\sin\omega(\tfrac{\pi}{\omega}) - \omega\cos\omega(\tfrac{\pi}{\omega})\right] - 1(-s\sin 0 - \omega\cos 0)\right]$$

$\int e^{ax}\sin bx\,dx = e^{ax}\left[\frac{a\sin bx - b\cos bx}{a^2+b^2}\right]$



$$\frac{\pm w\left(e^{-\frac{s\pi}{w}}+1\right)}{\left(1+e^{-\frac{s\pi}{w}}\right)\left(1-e^{-\frac{s\pi}{w}}\right)(s^2+w^2)} = \frac{1-e^{-\frac{2\pi s}{w}}}{1-\left(e^{-\frac{\pi s}{w}}\right)^2}$$